From EEG Intent to Simulated Robotic Pickup: A Midterm Project Reflection

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Abstract: This project investigates whether a low-bandwidth motor-imagery electroencephalography (EEG) interface can reliably enable and pause a learned simulated robotic pickup skill. Our initial proposal targeted the evaluation gap between offline neural classification and physical task execution: an above-chance window accuracy does not establish that noisy, flickering predictions can drive contact-rich manipulation within a fixed time budget without unintended motion during rest. We decomposed the system into per-person rest-versus-hand-imagery decoding, causal streaming replay with command gating, and an autonomous manipulation policy wrapped in a project-built `HOLD` and completion-latch adapter. Across a corrected five-person scripted Franka benchmark (15 episodes) and 36 rollouts coupling recorded EEG to an attributed pretrained behavior-cloning recurrent policy (`BC-RNN`), the pipeline achieved 21/27 retained lifts on three held-out participants (`S001`–`S003`) and 3/9 on an exploratory participant (`S006`). Crucially, physical outcomes separated into two failure regimes governed by an empirical minimum activation threshold of $(59, 120]$ in-window commands: strict gates starved recall-limited decoders (`S002`: 3/3 raw vs. 0/3 filtered), whereas precision-limited decoders completed 9/9 rollouts by false-triggering across 93.3% of rest windows (`S003`). For the remainder of the semester, we drop custom from-scratch reinforcement learning to execute a frozen, matched-control study on uninspected participants (`S011`–`S015`) pairing task completion with rest-motion and calibration-bounded gate evaluation.

1 Introduction

Non-invasive EEG offers an accessible assistive command channel, yet motor-imagery signals exhibit low signal-to-noise ratios, nonstationarity, and pronounced cross-subject variability [1, 2]. Standard BCI benchmarks demonstrate that offline accuracy frequently fails to transfer across splitting choices or cohorts [3], and an isolated classifier score leaves a critical robotics question unanswered: *when predictions stream causally into a stateful controller, do stabilization filters improve task completion, or do they starve the robot of the sustained activation required to grasp?* Our original twelve-week proposal therefore aimed to build a reproducible, simulation-first evaluation harness connecting public EEG to simulated robotic manipulation rather than claiming a live or clinical device.

Original gap, insight, and plan. The proposal identified that training an end-to-end policy directly from noisy EEG would compound neural nonstationarity with contact discovery. Our core insight was hierarchical shared autonomy [4]: restrict the decoder to a conservative binary choice—instructed rest (`T0`) versus combined left/right fist imagery (`T1`/`T2`) in the PhysioNet EEG Motor Movement/Imagery dataset [5]—pass streaming probability estimates through a command gate (`ENABLE` vs. `HOLD`), and delegate continuous joint control to a low-level policy. Mapping fist imagery to a pickup request is an explicit interface convention, not evidence that the dataset encodes native grasp intent. Our twelve-week schedule planned leakage-safe Common Spatial Patterns (CSP) [6], Riemannian [7], and EEGNet baselines (Weeks 1–6), causal replay and temporal command filtering (Weeks 7–8),

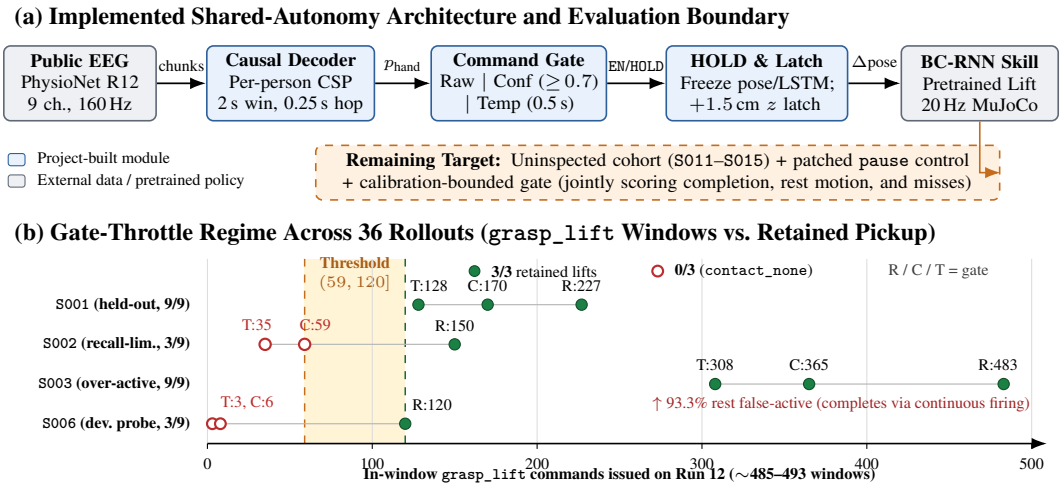


Figure 1: **System overview and midterm empirical regime split. (a)** Project-built modules (blue) connect public motor-imagery EEG to an externally pretrained `robomimic` `BC-RNN` policy (gray) via causal decoding, command gating, and a `HOLD`-preserving completion latch; orange indicates the remaining prospective benchmark. **(b)** Across 36 learned-policy rollouts (3 scene seeds per gate), cube acquisition and terminal retention separate cleanly across an empirical command-count threshold of $(59, 120]$ `grasp_lift` windows, exposing a tradeoff between strict-gate starvation (`S002`, `S006`) and unbounded rest false activation (`S003`).

Isaac Lab reinforcement learning (RL) pickup training (Weeks 9–10), and matched end-to-end trials (Weeks 11–12). Figure 1 illustrates the implemented architecture alongside our central empirical finding at midterm.

2 Reflection: Where are we?

EEG decoding and scripted baseline. We built a leakage-safe MNE preprocessing and causal replay pipeline calibrating per-person models on runs 4 and 8 with run 12 held out (9 sensorimotor channels, 8–30 Hz, 2.0 s windows). On participants `S001`–`S005` (150 test windows), individual CSP/LDA reached 70.67% balanced accuracy (29.33% rest false activation), whereas subject-pooled CSP/LDA dropped to 49.33%; Riemannian tangent-space classification reached 72.67% individual and 70.00% pooled accuracy. On a separate five-person cohort (`S006`–`S010`), individual CSP/LDA scored 59.33% balanced accuracy (29.33% rest false activation). Driving a frozen *scripted* Franka lift over 120-second replays after fixing an initial `HOLD`-drift bug, raw, confidence (≥ 0.7), and temporal (0.5 s persistence) gates achieved physical lifts in 5/5, 3/5, and 2/5 subjects, respectively. Filtering reduced mean rest command activation from 38.7% to 25.7% and 19.6% (and physical rest motion from 13.5% to 9.1%), but increased missed imagery bouts from 12 to 30 and 38.

Robot learning progression and iterative latch engineering. Custom policy learning in Isaac Lab proved substantially harder than anticipated. While our PPO agent lifted from privileged in-hand resets, ready-pose table transfer yielded 0/5 pickups; curriculum refinements reached 2/5 lifts in a near-table progress screen (with one drop and 92–169 N asymmetric right-finger impacts) and peaked at 3/5 sustained bilateral acquisitions in a unilateral near-table screen, without ready-pose reliability or `HOLD` invariance. We therefore pivoted to an attributed, externally pretrained `robomimic` `BC-RNN` Panda Lift policy [8] in native MuJoCo (archiving 11.9 GB of simulation checkpoints and traces off-host). Even uninterrupted 400-step native rollouts lifted briefly yet retained the cube at step 400 in 0/5 runs. Across four controlled adapter phases, freezing the end-effector pose target and LSTM state during `HOLD` yielded 3/21 terminal retentions; adding an irreversible grasp latch raised retention to 17/21 (four sub-centimeter height dips to 7.72–7.98 cm); and applying a one-time $+1.5$ cm world-z

offset at latch acquisition recovered all four failures to reach **21/21** retained positive controls and 0/3 on `always-hold` negatives.

Connected EEG-to-learned-policy results (Nov. 4–6). Using this frozen Phase 6 regime, we executed 36 offline decision-stream rollouts (400 steps / 20 s at 20 Hz; seeds 0–2 per gate) across robot-development subject `S006` and all three subjects held out from adapter tuning (`S001`–`S003`, each per-person calibrated on runs 4/8 and tested on held-out run 12), bracketed on `S001` by matched `always-enable` (3/3 retained, steps 47–58) and `always-hold` (0/3) controls:

- **Balanced (`S001`, 76.7% BA, 20.0% rest false-active):** Completed and retained **9/9** lifts across raw (227 commands), confidence (170), and temporal (128) gates with zero drops.
- **Recall-limited (`S002`, 73.3% BA, 53.3% hand recall; and dev. `S006`):** Completed **3/9** each—retaining 3/3 under raw commands (`S002`: 150; `S006`: 120) but 0/3 under confidence (`S002`: 59; `S006`: 6) and temporal (`S002`: 35; `S006`: 3) gates, which starved the policy of reach time.
- **Precision-limited (`S003`, 53.3% BA, 93.3% rest false-active):** Completed **9/9** lifts across all three gates (483/365/308 commands), succeeding physically because near-continuous false firing acted like an always-on trigger.

Was the gap real, did we verify our insight, and what was harder than expected? The gap between offline decoding and closed-loop utility proved sharper than hypothesized, though as an empirical systems bottleneck rather than an unstudied paradigm [4, 9]. Uncalibrated LDA posteriors caused fixed-threshold (≥ 0.7) duty cycles to swing nearly $50\times$ across people (0.012 on `S006` vs. 0.565 on `S008`), making accuracy a poor proxy for control: `S002` (73.3% BA) failed strict gates, whereas `S003` (53.3% BA) passed 9/9 via unintended rest activation. Our shared-autonomy insight is verified in bounded form—binary EEG replay can drive a learned pickup skill once mediated by a recurrent-state-preserving `HOLD` and grasp latch—while falsifying our assumption that fixed temporal filtering uniformly helps. Harder-than-expected challenges included terminal grasp equilibrium, preventing `HOLD` target drift, and auditing control code—where our Nov. 6 review caught and patched an inverted `pause`-matched control branch (`hold` vs. `policy` inside the timed window) before any invalid causal bottleneck localization was claimed.

3 Updated work plan

To guarantee a rigorous end-of-semester contribution, we retire from-scratch Isaac Lab RL and focus our remaining compute (CPU MuJoCo plus GPU acceleration) on the brain-to-robot gating bottleneck. Alongside prospective evaluation on genuinely uninspected subjects, we introduce a *conformal risk-controlling gate* calibrated strictly on runs 4/8 (with optional cross-subject Riemannian/EEGNet pretraining using GPU acceleration) to statistically bound rest false activation without starving recall-limited users.

Table 1: Revisions from the initial twelve-week proposal to the final semester plan.

Original component	**Status**	**Revision and technical rationale**
EEG baselines & replay	Complete	Per-person/pooled CSP and Riemannian pipelines plus causal replay verified; retain binary `T0` vs. `T1`/`T2` labels.
Isaac Lab PPO table pickup	Stopped	Peaked at 3/5 near-table lifts with asymmetric finger impacts; archive 198 checkpoints and use attributed `BC-RNN`.
`HOLD` & retention adapter	Complete	Progressed $0/5 \to 3/21 \to 17/21 \to 21/21$ via state-freezing `HOLD` and $+1.5$ cm latch; patched `pause`-control inversion.
Learned-policy EEG replay	36 runs done	24/36 retained lifts (`S001`–`S003`, `S006`) exposed the $(59, 120]$ trigger threshold; test unopened `S011`–`S015`.
Model & gate scope	Refocused	Defer unsupported wrist/twist labels; use GPU acceleration for Phase C (CSP/Riemannian/EEGNet) and conformal risk gating.

Remaining schedule (Weeks of Nov. 9 – Dec. 7). (1) Nov. 9: Run the patched `pause`-matched control batch on `S001`–`S003` to separate duty-cycle sufficiency from fragmented EEG timing, run the Phase C decoder comparison, and freeze conformal calibration rules (fit on runs 4/8 only) plus unseen scene seeds. **(2) Nov. 16–23:** Execute one locked prospective evaluation on five completely uninspected participants (`S011`–`S015`, run 12) comparing raw, fixed-confidence, temporal, and conformal risk-bounded gates against `always-hold`, `always-enable`, and duration-matched `pause` controls. **(3) Nov. 30:** Join cue annotations to MuJoCo telemetry to jointly score retained lifts, robot motion during annotated rest, missed bouts, and latency—testing whether conformal gating suppresses `S003`-style rest firing while avoiding `S002`-style starvation. **(4) Dec. 7:** Package the final paper, synchronized EEG-and-MuJoCo replay video, and reproducible scripts, reporting any negative outcomes directly without relaxing thresholds or claiming live closed-loop BCI validation.

4 AI statement

AI coding and writing assistants (OpenAI Codex, Claude, and Google Antigravity) were used to help organize literature metadata, scaffold simulation/evaluation scripts, audit control harnesses, summarize logged telemetry, and refine the prose and TikZ figure in this report. All numerical claims, SHA-256 artifact records, and experimental interpretations were verified against saved repository outputs, externally pretrained models are explicitly attributed, and full responsibility for the technical conclusions and limitations rests with the author.

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